



Society of Petroleum Engineers

SPE-227679-MS

Laboratory Investigation of CO₂ Trapping in Carbonates Aquifers – The role of Micropores

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Carbonate rocks have complex pore systems and their suitability for CO₂ storage is contingent on favorable pore structure. Unfortunately, there is insufficient data about the screening and selection process for gas storage in carbonate rocks. To test the suitability of some carbonate core samples for storage, we carried out pore analysis of the core samples, which included helium porosity/permeability, capillary pressure, and NMR relaxation test. A porous plate drainage experiment at a capillary pressure of 200 psi followed by NMR measurements were conducted to determine the T₂ cut-off of the pores. CO₂ coreflood experiments were also carried out at high pressure and temperature to determine the storage efficiency of the rock. The overall results show that the presence of intragranular pores can significantly hamper storage capacity, especially if they form a significant portion of the pore systems. High T₂ cut off values ($\approx$ 200 ms) were found in the cores, where the intraparticle pores have high capillary entry pressure ($>$ 200 psi). Analysis of initial-residual saturation from coreflooding data showed that the core samples have a trapping efficiency ranging between 1.8 to 2.2. However, further observation revealed that only the intergranular pores were available for CO₂ storage, and only a fraction of these pores was drained by the injected CO₂. In some carbonate rocks, the intragranular pores constitute about half of the total pore system which further reduces the storage capacity of the rock. These observations highlight the importance of a well-defined screening criteria for gas storage in carbonate rocks. The fraction of intragranular to intergranular pores and the T₂ cut-off should be considered as one of the most important screening criteria in carbonate rocks.

Keywords: CO₂ storage, Saline Aquifer, Carbonate rocks, Site screening and selection, Intraparticle pores

Introduction

Geological candidate sites for CO₂ sequestration include saline aquifers, depleted oil and gas reservoirs, coal beds, and potentially basalt formations [1, 2]. In these, CO₂ is either physically trapped in the pores

or chemically trapped after a reaction between dissolved CO₂ and rock minerals, resulting in carbonate minerals precipitating. Carbon storage in deep saline aquifers has attracted significant attention due to their high capacity for CO₂ storage, among other options [3]. Currently, depleted oil reservoirs in Saudi Arabia are extremely scarce to nonexistent, and the situation will remain as such for many decades. Hence, CO₂ storage in depleted aquifers is not feasible in the Kingdom now. When injected into an aquifer, CO₂ migrates to the top of the reservoir beneath the caprock [4]. The amount of CO₂ that could be stored in saline aquifers for a reasonable amount of time is large. For example, the CO₂ storage capacity of the Alberta deep saline basin was estimated at 103 Gt, and the injection of CO₂ into the Utsira aquifer in the North Sea amounted to >16 Mt [5].

Saudi Arabia is abundant in deep and shallow aquifers suitable for CO₂ sequestration, which can be found in both the stacked sedimentary units of Eastern Arabia and the Western rift basins [6, 7]. Furthermore, the proximity of CO₂ emission sources and sinks in Saudi Arabia renders it a unique economic advantage considering the cost of CO₂ transportation [8]. Since carbonate reservoir formations form a significant percentage of the world's subsurface reservoir rocks [9, 10], it is important to investigate the role of the complex pore structure of carbonate rocks in the storage efficiency of CO₂. This study examines the role of the complex pore structure of carbonate rock on CO₂ storage efficiency.

Methodology

Sample preparation

Two carbonate core samples named sample #1 and sample #2 were cleaned, dried at 60 °C in a vacuum oven, and the sample dimensions were measured. This was followed by measuring the gas permeability and porosity for all the samples. The samples were vacuumed for two days and saturated with brine (60,000 ppm, see Table 1) under high pressure. At 100% water saturated state, the T₂ nuclear magnetic resonance (NMR) distributions were conducted on all the samples to obtain their pore size distribution. Table 2 shows the properties of the samples used in the core flooding tests, which include the dimensions, porosity, permeability, and T₂ cut-off of the rock samples. The capillary pressures in the samples were also conducted using a porous plate experiment, where humidified nitrogen was used to desaturate the core samples using different capillary pressure steps until irreducible water saturation was attained. The weights of the samples were taken at the different capillary pressures to evaluate the saturation at each capillary pressure. This allowed the capillary pressure versus saturation curve to be generated for the two samples.

Table 1—Salt composition per Liter of DI water.

Salt type	Concentration, g/L (60,000 ppm)
NaHCO ₃	0.14
Na ₂ SO ₄	0.15
NaCl	42.23
CaCl ₂	14.81
MgCl ₂	2.68

Table 2—Properties of rock samples

Salt type	Length (cm)	Diameter (cm)	He Porosity (%)	Gas Perm. (mD)	T ₂ cut-off (ms)	Irreducible water (dec.)
Sample #1	5.64	3.784	13.10	251	200	0.21
Sample #2	9.97	3.801	19.53	200	200	0.47

Coreflood procedure

A clean and dry core sample was placed inside the core holder in a coreflood apparatus shown in Fig. 1, the core was flushed with dry CO₂ at ambient conditions while an overburden pressure was applied. Based on the designed experiment, two HPHT vessels were filled with CO₂ gas and the prepared synthetic brine. The rock was pressurized with carbon dioxide gas at experimental pressures while maintaining a net overburden pressure (NOP) of 800 psi (i.e., NOP = overburden pressure – pore pressure). The system was heated to 60 °C in all the tests. At this stage, the core sample was scanned to obtain a background scan of the gas, i.e., N₂ or CO₂, saturated sample. After obtaining the gas background scan, the core sample was depressurized to ambient pressure and vacuumed for 6 hours. The negative pressure in the sample (caused by vacuum) allowed the synthetic brine to saturate the sample. This was followed by pumping more brine into the sample under a net overburden pressure of 800 psi until several pore volumes were injected and the core sample was fully saturated. The core sample was then further pressurized by injecting brine at low injection rates (0.5 cc/min) while increasing the back pressure until it reached the target value of about 1450 psi. A brine background scan was performed at this stage. For the CO₂ Initial-Residual (IR) test, the dead brine was displaced with the pre-prepared live brine, and another background scan was performed at experimental conditions, which is needed for non-wetting phase (gas) saturation calculation. It should be noted that the live brine was equilibrated with the rock before being injected into the core sample to avoid any reactions. Thereafter, residual trapping (*IR*) tests were performed. Initially, live brine was injected into a core fully saturated with brine. This was followed by the primary drainage (gas) injection until the maximum gas saturation was reached. The core was scanned to measure the maximum or initial gas saturation (S_{gi}) along the core length. The last stage was the secondary imbibition of the live brine to displace the gas out of the core, while the remaining gas saturation that could not be displaced is named "residual gas saturation" (S_{gr}). The core was scanned again at the end of this stage to measure the residual gas saturation along the core length. It should be noted that the previously described cycle was repeated with various primary drainage injection rates, ranging from 0.2 cc/min to 7 cc/min, to increase the initial gas saturation along the core. Hence, the corresponding residual gas saturation profile was changed based on the initial saturation range before the secondary imbibition process. For all cycles, the imbibition was performed at a constant flow rate of 0.2 cc/min.

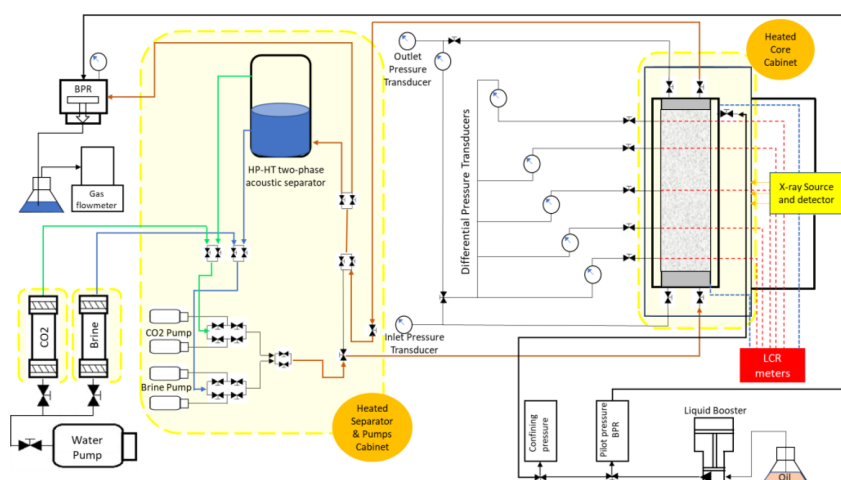


Figure 1—Coreflood apparatus

Results and discussions

The pore size distribution of the two samples are represented by the capillary pressure curve and NMR T_2 distributions shown in Fig. 2 (for sample #1) and Fig. 3 (sample #2). As seen in the figures, the irreducible

water saturation is quite low for sample #1 ($S_{wir} = 0.21$) while that of sample #2 is quite high ($S_{wir} = 0.47$). NMR T_2 have a one to one correlation with pore sizes. The bigger the small size the higher the T_2 values and vice versa. Hence, a probability density function of T_2 can give an indication of the distribution of pore sizes in a geological rock. It is also important to note that NMR signals only come from the water filling the respective pores. When the water is no longer in any of the pores, the magnitude of their T_2 values becomes zero. Hence, for a complete characterization of the entire pore system of a rock sample, the sample has to be fully saturated with water before NMR measurements. The NMR T_2 distributions in Fig. 2 (for sample #1) and Fig. 3 (sample #2) show that the two samples have bimodal pore systems, i.e. the presence of micropores (intra granular pores) and macro pores (inter granular pores). The thick continuous curves in the NMR figure represent the probability density function of the T_2 values analogous to the pore size distribution in MICP data. The broken lines represent the cumulative sizes of the pores. As seen in both figures, Fig. 2 and Fig. 3, the two samples are quite identical. They are both carbonate rocks with similar permeability (251 mD and 200 mD). They both have two pore systems. They both also have similar T_2 cut-off value of ≈ 200 ms, which is about twice the common values (≈ 100 ms) for many carbonate rocks. The T_2 cut-off values were obtained after desaturating the cores by a capillary pressure of 100psi. However, the ratio of the micropores to macro pores are quite different for the two samples. The micropores constitutes 25% of the total porosity in sample #1 and 48% in sample #2, while the macropores constitute 75% and 52% respectively. Fig. 4 is a thin section image showing the micro and macro pores. The significant volume of micropores in sample #2 is responsible for the high irreducible water saturation as a high amount of water is trapped in these micropores and could not be displaced by the injected gas during the porous plate capillary experiment. Coreflood experiments on the two samples was used to evaluate and compare the storage and trapping efficiency of CO_2 in the two samples.

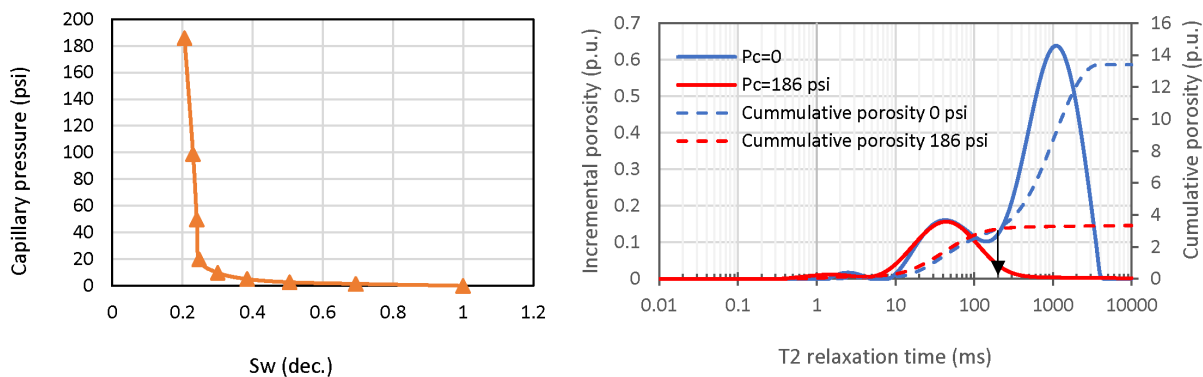


Figure 2—capillary pressure curve (a) and NMR T_2 distribution (b) of sample #1

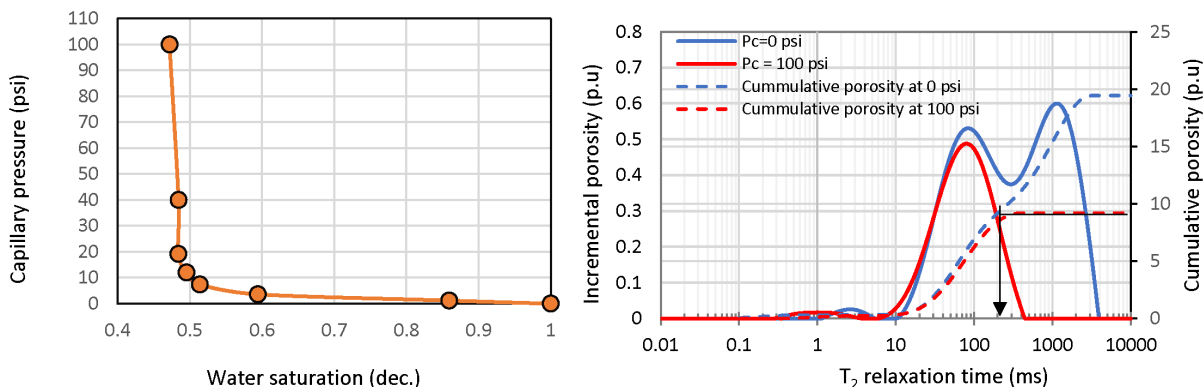


Figure 3—capillary pressure curve (a) and NMR T_2 distribution (b) of sample #2

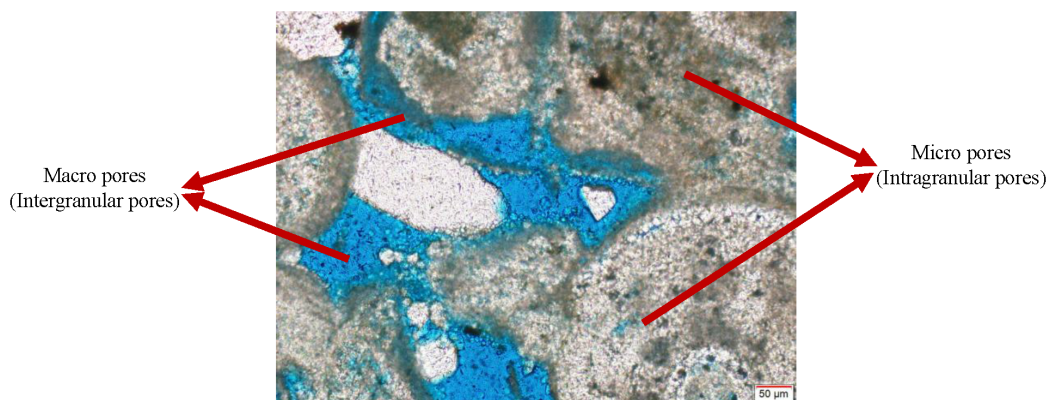


Figure 4—Thin section microscopy of sample #2 showcasing the two pore systems.

The initial residual saturation experimental data are shown by the data points in Fig. 5 and Fig. 6. The data was fitted with Land's model [11] shown in Equation (1). The model describes the residual trapping characteristic of a porous medium by relating the residual trapping saturation (S_{gr}) of the nonwetting phase to the maximum gas saturation (S_{gi}) obtained before imbibition. The parameter C was obtained by fitting Equation 1 to the data points. A 'zero' Land coefficient ($C = 0$), represented by the black dotted unity line, means that all the injected gas is trapped by capillary forces during water imbibition. Hence, the higher the land coefficient, the lower the trapping efficiency of the rock. Sample #2 has a Land coefficient of 2.2, which is much higher than that of sample #1 (with $C = 1.8$). Obviously, the relatively poorer trapping efficiency of sample #2 compared to sample #1 is due to the significant fraction of the micropores.

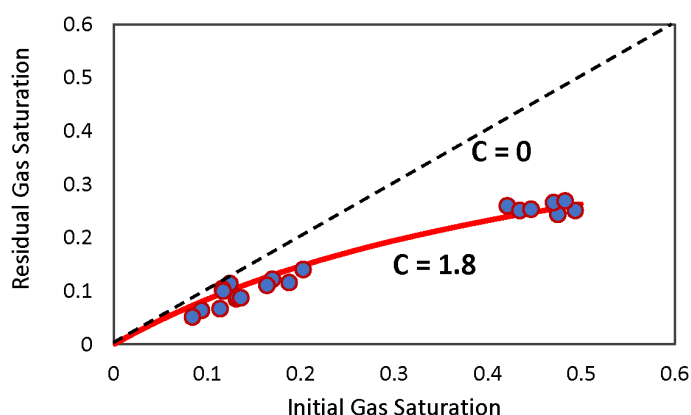


Figure 5—Initial-residual curve for sample #1

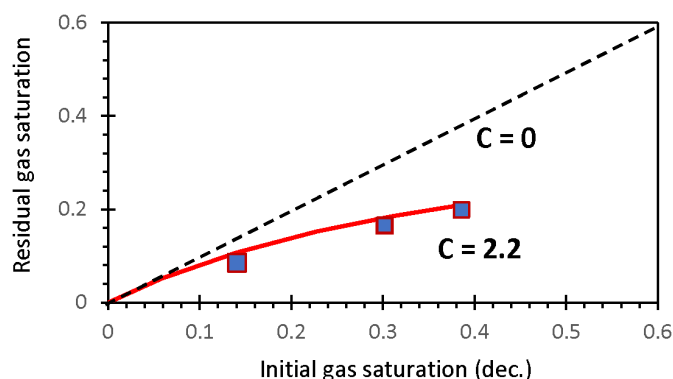


Figure 6—Initial-residual curve for sample #2

$$S_{gr} = \frac{S_{gi}}{1 + CS_{gi}} \quad (1)$$

The fitting was done by adjusting the value of the trapping coefficient (C) such that a minimum objective function (E) was attained. The objective function is represented by the equation (2), which represents the difference between the values of the measured residual gas saturation ($S_{r_{Measured}}$) and the calculated residual gas saturation, using Equation (1).

$$E = \sqrt{\frac{\sum_{i=1}^N (S_{r_{Land's}} - S_{r_{Measured}})^2}{N}} \quad (2)$$

where $i = 1, 2, 3, \dots, N$, are the saturation points along the core length.

The maximum CO_2 injected in each sample and the corresponding residually trapped CO_2 are summarized in Table 3. NMR measurement on sample #2 after CO_2 injection showed that the injected CO_2 occupies only the macropores, while the micropores remain uninvaded as shown in Fig. 7. The blue curve shows the distribution of the water in the pores when the sample was fully saturated with water (i.e. $S_w = 100\%$) as also shown in Fig. 3. After CO_2 injection, the water in the macropores were drained. As a result, the magnitude of the T_2 distribution in those pores reduced to zero as shown by the green curves in Fig. 7. The injected CO_2 could not displace the water in the micro pores because of the high entry capillary pressure of that pore system. The NMR measurements was not conducted on sample #1 after CO_2 injection, but the insight obtained from sample #2 also applies to it. The entry capillary pressure of the micropores have to be determined at higher experimental capillary pressure. From Fig. 3., it appears to be somewhere above 200psi. CO_2 storage may be challenging in rocks with a significant fraction of micropores similar to sample #2. Innovative techniques are needed to divert or force the injected CO_2 into those micropores, so as to optimize the storage efficiency of such rocks. This is an area needing active research since carbonate rocks are abundant and are inherently bimodal in nature. Few studies have investigated the application of foam injection technique in bimodal carbonate systems [12, 13].

Table 3—Summary of gas trapping

	Maximum injected CO_2 (%*)	Residually trapped CO_2 (%*)	Land's trapping coefficient (C)
Sample #1	49	27	1.8
Sample #2	38.6	19.8	2.2

*

% here mean the percentage of the total pore volume occupied by CO_2 (also referred to as saturation)

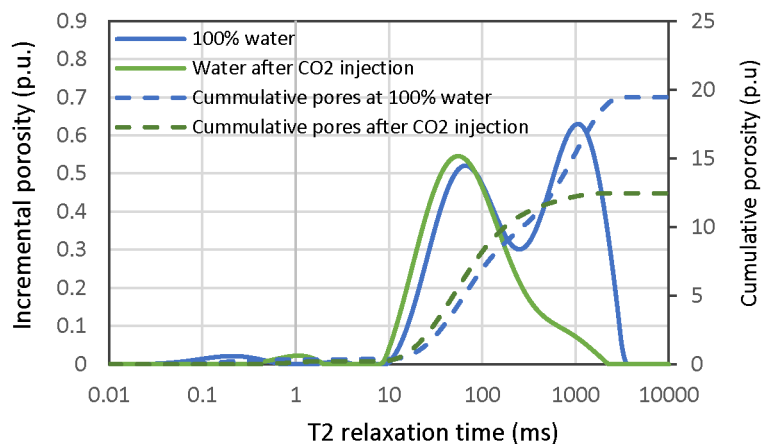


Figure 7—NMR T_2 distribution after CO_2 injection in sample #2

Conclusions

The following conclusions are made from this work:

1. Carbonate rocks can have a significant fraction of micropores that constitutes a large portion of the total porosity of the rock.
2. The T_2 cut-off of carbonate rock that delineate the micro and macro pores can also be as high as 200ms or more.
3. Carbonate rocks are likely to offer only the macro (intergranular pores) pores for CO_2 storage as the micropores retain the connate water at high entry capillary pressure, which can be in excess of 200psi.
4. For carbonate rocks, the high values of effective porosity and permeability alone may not sufficient indices to determine a good rock candidate for CO_2 storage. There is a need to incorporate the ratio of micropores to macropores in the screening criteria for carbonate rocks.
5. NMR T_2 cut-off (for micropores) should also be considered as a screening criterion in carbonate rocks.

Acknowledgement

The authors are grateful to Saudi Aramco for sponsoring this work through a research grant CIPR 2358.

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